UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which statement immediately exits a loop?
   A. cin
   B. return 0
   C. continue
   D. break

2. What is displayed by the following code?

```
int a[4] = {3, 6, 9, 12};
cout << a[1] + a[3];
```

   A. 9
   B. 18
   C. 12
   D. 15

3. How many times does a while loop execute when its condition is false initially?
   A. Until the user stops it
   B. One
   C. Zero
   D. Two

4. Which structure chooses between two alternative paths?
   A. if...else
   B. struct
   C. array
   D. for

5. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. i and ii
   B. iii and iv
   C. ii and iv
   D. i and iii

6. Which values can a bool variable hold?
   A. any decimal
   B. any character
   C. any integer
   D. true or false

7. Which term describes a finite sequence of precise steps that solves a problem?
   A. Algorithm
   B. Array
   C. Compiler
   D. Variable

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8. Which item is NOT a programming paradigm?
   A. Procedural programming
   B. Object-oriented programming
   C. Debugging
   D. Structured programming

9. Which identifier is valid in C++?
   A. 2total
   B. int
   C. total score
   D. total_score

10. Which input pattern stops when the user enters -1?
   A. A function prototype
   B. A sentinel-controlled loop
   C. A fixed for loop
   D. A switch statement

PART B (60 MARKS)

QUESTION 1

This question tests programming concepts and error terminology.

a) Define compiler and interpreter.

(3 marks)

b) State THREE differences between a compiler and an interpreter.

(2 marks)

QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

Table 1. Concert Ticket Prices

Age Range	Ticket Price (RM)
0 - 12	15.00
13 - 17	25.00
18 and above	40.00

a) Draw a flowchart for the ticket price selection.

(6 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(7 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(5 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
    s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(10 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(5 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A cafe sells the items shown in Table 1. A 10% discount is given when the subtotal is at least RM50.

Table 3. Cafe Menu

Item	Price (RM)
Coffee	8.50
Tea	6.00
Juice	9.50

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

QUESTION 2

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

END OF QUESTION PAPER

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